1 OBJECT

This document describes the data format and protocol for AL22 when it communicates with a LIS system. The functionalities have to match the software specifications.

2 APPROVAL AND HISTORY

Révision	Date	Auteur	Raison et nature de la modification	Approbation
01	20/09/10	V.Heaney	Création du document	J. Stawicki
02	15/02/11	V.Heaney	Translation in English, suppress SVM, update for V2.0	J. Stawicki

3 DIFFUSION / MISE A DISPOSITION

4 TERMINOLOGY AND ABBREVIATIONS

Host	laboratory informatic system, HOST system
CT	Closed tube on racks
OV	Other vial (open tube)
Hand-Shake	Bi-directionnnal communication using acknowledgement.

5 REFERENCES

Format description uses following symbols :

Symbole	**Valeur Hexadécimal ou signification**
[CR]	0x0d, carriage return
[LF]	0x0a, line feed
[CRLF]	0x0d 0x0a
[TAB]	0x09, tabulation
[SPACE]	0x20, espace

[ETX]	0x03, end of text
‘A’	0x41 , a character is within quotes
(OPT)	Optional field . If not received, Mythic22AL will take a default value
(SEND)	The following frame is sent by **Mythic22AL**.
(RECEIVE)	The following frame is received by **Mythic22AL** .

Hand-Shake : communication bidirectionnelle basée sur l’acquittement.

6 SYSTEM OVERVIEW

The Mythic22AL system may communicate with a LIS system. The essential part of these communications is the results sending.

7 GENERAL SPECIFICATIONS

It is a specific format designed by C2 to transfer data to an external system using RS232 or Ethernet interface. The setting and wiring are not described in this document.
To make easy the use of these data, the chosen format is text line oriented and is compatible with CSV format. The identifiers (key word) have to be in English.

7.1 Generic principles

The exchanged **frames** are in ASCII code.
A frame contents one or more **lines**.
A line contents one or more **fields**.
All sended lines by MythicAL22 are terminated with [CR], even if not specified in the examples.
In receiving, MythicAL22 can accept [CR], [LF] or [CRLF] as end line.

Le Mythic22AL begins each frame by the following header :

22AL X ;Y ;Z(; status parameters if any)[CR]

where :
X is the instrument **number of** the Mythic22AL (max 2 chars).
Y is the user identifier (login) (max 10 chars).
Z is the frame identifier
The decimal separator is the dot.
The field separator is the semicolon.
The DATE format is set to DD/MM/YYYY, time is set to HH:MM:SS in 24 hours mode.

The atomic frames are:

- Results sending request with results sending
- Sending of calibration packet with calibration results

All the frames lines have to be emitted (if there is no information, the parameter is left empty)..
All the identifier key words (of frame or of parameter) are in upper-case.
The following fields (SID, PID, ID, blood type, operator, lot number) are UTF8 encoded in UTF8.

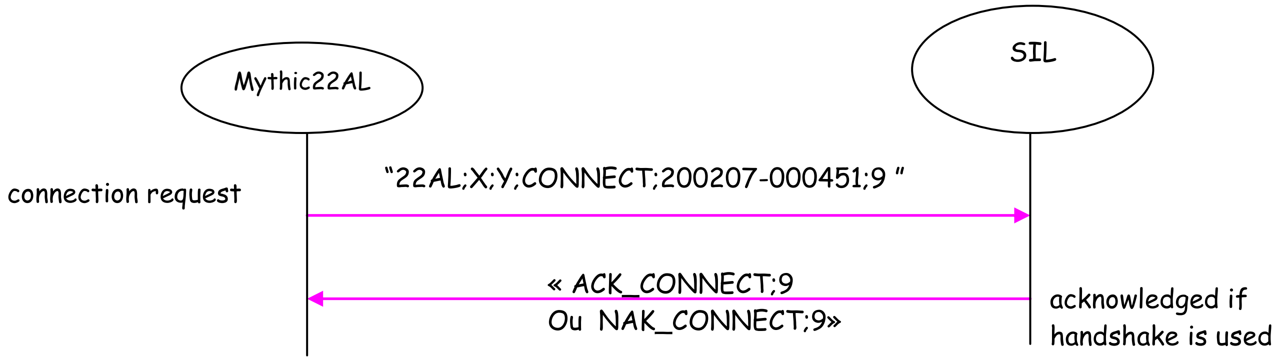
Maximum fields length for fields shared among different frames	
OPERATOR	10 chars
USER	10
LOT (CALI ou QC)	8
SID	16
PID	16
ID	20

7.2 FRAMES

7.2.1 Connection

To communicate with a Host, it is not required to establish this connection. However, it is available to secure the communication.

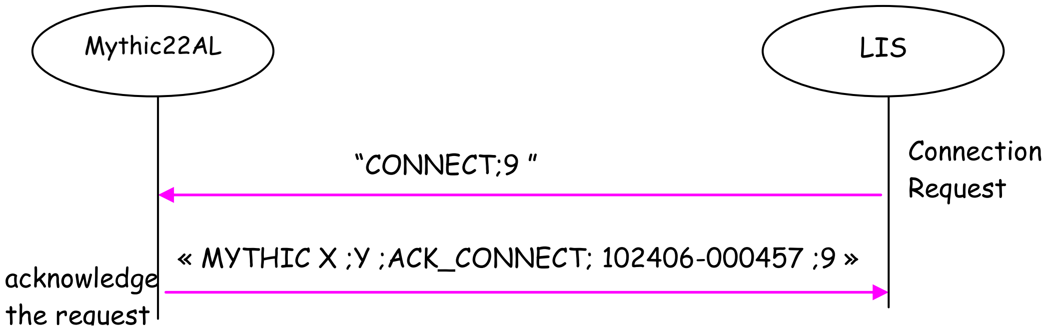
7.2.1.1 Mythic22AL connection resquest



(SEND)
22AL X ;Y ;CONNECT ;Serial number (*);C2 format version(**)[CR]
Ex : 22AL 2 ;GG ;CONNECT ;102406-000457 ;9
ACK_CONNECT ; C2 format version [CR]
And if the instrument serial number is not allowed to connect:
(RECEIVE)
NACK_CONNECT ; C2 format version [CR]

In the case of an unvalid motherboard serial number (not numeric) then the serial number is transmitted entirely.
(**) This will remain to 9 to do not break compatibility with actual connections.

7.2.1.2 LIS connection request



The LIS sends a frame :
(RECEIVE)
CONNECT ; C2 format version [CR]

Mythic22AL acknowledges :
(SEND)
22AL X ;Y ;ACK_CONNECT ; Serial number (*) ;C2 format version [CR]

7.2.1.3 Units

The parameters unit is specified by the line :
UNIT ;u[CR] where u is the unit code.
This line is used for host connection.

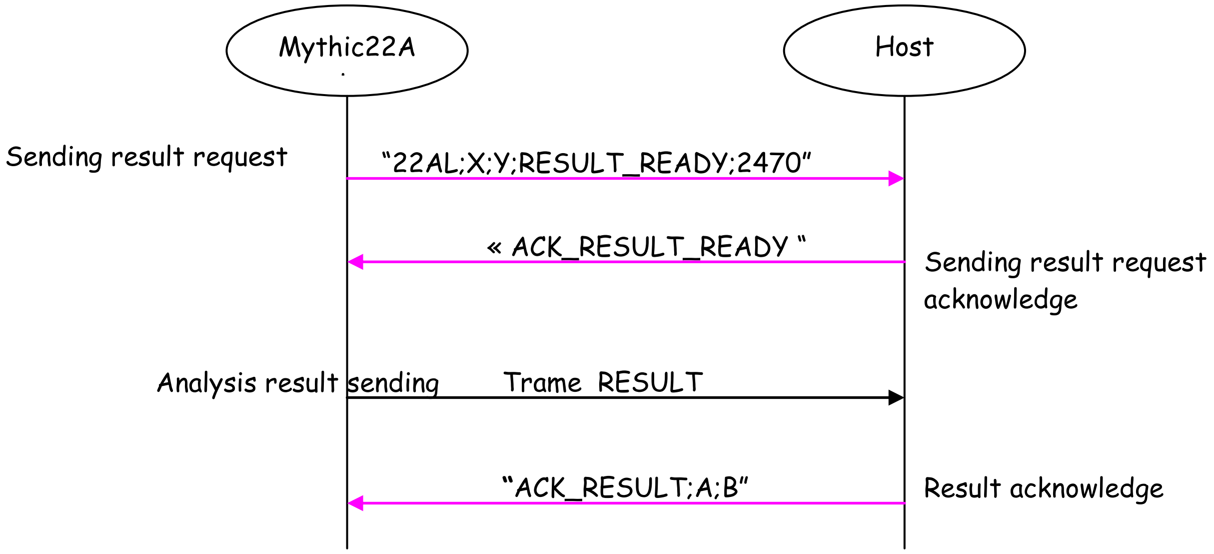
- Host Mode: communication are done with the current unit set for the instrument.

Unit code value	Unit system
1	STANDARD
2	S.I.
3	S.I. MMOL

If a received frame does not specify the unit code, then the Mythic22AL will use STANDARD unit by default.

7.2.2 RESULT (SAMPLE)

The Mythic22AL sends the results after the Host acknowledge occurs. This behaviour avoids a stream to be sent with no host connected.



HOST with hand-shake mode (time-out can be set)

7.2.2.1 Sendind result request

Utilisée dans le cas de fonctionnement en mode « Hand-Shake »
Used in the case of a hand-shake mode.
(SEND)
22AL X ;Y ;RESULT_READY ;Taille[CR]

7.2.2.2 Sendind result request acknowledge

(RECEIVE)
ACK_RESULT_READY[CR]

7.2.2.3 Analysis result sending

This concerns results which are not linked with calibration, quality control or repeatibility.

(SEND)
22AL X ;Y ;**RESULT** [CR]

DATE;30/10/2003 [CR]
TIME;15:36:38 [CR]
MODE;**NORMAL** [CR]
UNIT ; Unité [CR] specify the unit used for the following parameters
SEQ;352; [CR]
SID;3 [CR]
PID;X28 [CR]
ID;DUPONT [CR]
TYPE;STANDARD [CR]
TEST; CBC ou DIF [CR]
-------- With the M22 RESULT FORMAT unchecked, the following fields are included :
RTYPE;1 [CR]
RACK;2 [CR]
POS;5 [CR]
BIRTH;14/09/1981 [CR]
SEX;1 [CR]
PRESC;DR HOUSE [CR]
LOCAT;PARIS [CR]
~~FIRSTNAME;VINCE [CR]~~
DRAW DATE; (0 for UNKNOWN ou 1 for TODAY 2 for YESTERDAY)
DRAW TIME; hh:mm:ss
PATIENT COMMENT;NO COMMENT [CR]
INFO;;;

OPERATOR ; Mythic22AL login at run time[CR]
PREL;CT [CR]
CYCLE;A [CR]
WBC; 11.0;A;B; 2.0; 4.0; 11.0; 15.0 [CR]
RBC; 6.00;A;B; 2.50; 4.00; 6.20; 7.00 [CR]
HGB;15.0;A;B; 8.5;11.0;18.8;20.0 [CR]
HCT;49.0;A;B;25.0;35.0;55.0;60.0 [CR]
PLT; 320;A;B; 70; 150; 400; 500 [CR]
LYM; 3.4;A;B; 0.8; 1.0; 5.0; 5.5 [CR]
MON; 1.4;A;B; 0.0; 0.1; 1.0; 1.1 [CR]
LYM%;30.5;A;B;15.0;25.0;50.0;55.0 [CR]
MON%;13.0;A;B; 1.0; 2.0;10.0;12.0 [CR]
MCV; 81.7;A;B; 70.0; 80.0;100.0;120.0 [CR]
MCH; 25.0;A;B; 25.0; 26.0; 34.0; 35.0 [CR]
MCHC; 30.6;A;B; 28.0; 31.0; 35.0; 36.0 [CR]
RDW; 8.1;A;B; 7.0; 10.0; 20.0; 25.0 [CR]
MPV; 7.5;A;B; 5.0; 6.0; 10.0; 12.5 [CR]
PCT; 0.401;A;B; 0.150; 0.200; 0.500; 0.600 [CR]
PDW; 9.9;A;B; 5.0; 8.0; 18.0; 25.0 [CR]
EOS;1.4;A;B; 0.0;0.5;1.0; 5.0 [CR]
BAS;0.6;A;B; 0.0;0.5;1.0; 5.0 [CR]

EOS%;2.4;A;B; 0.0;0.5;1.0; 5.0 [CR]
BAS%;1.6;A;B; 0.0;0.5;1.0; 5.0 [CR]

WBC CURVE; valeur canal1;... ;valeur canal 128 [CR] histograms followed by thresholds
WBC THRESHOLDS;S1;S2 [CR]
RBC CURVE; valeur canal1;... ;valeur canal 128 [CR]
RBC THRESHOLDS; S1;S2 [CR]
PLT CURVE; valeur canal1;... ;valeur canal 128 [CR]
PLT THRESHOLDS; S1 [CR]
ALARMS;x;x;x;x;x;x;x;x;x;x;x [CR] alarms for the result
INTERPRETIVE_WBC;y;y;y;y;y;y;y;y;y;y;y;y [CR] Pathology message
INTERPRETIVE_RBC;y;y;y;y;y;y;y;y;y;y;y;y [CR]
INTERPRETIVE_PLT;y;y;y;y;y;y;y;y;y;y;y;y [CR]
LMNE MATRIX; [CR]
m;m;m;m;m;m;m;m;m;m;m;m;m m;m;m;m;m; [CR]
m;m;m;m;m;m;m;m;m;m;m;m;m;m;m;m;m;m;m;m;m m;m;m;m;m; [CR]
…………
m ;m ;m ;m ;m ;m ;m ;m ;m ;m ;m ;m ; [CR]
[CR]
LMNE SHADE MATRIX; [CR]
m;m;m;m;m;m;m;m;m;m;m;m;m m;m;m;m;m; [CR]
m;m;m;m;m;m;m;m;m;m;m m;m;m;m;m; [CR]
…………
m ;m ;m ;m ;m ;m ;m ;m ;m ;m ;m ;m ; [CR]
[CR]
THRES 5D LMNE MATRIX; [CR]
T ;S ;T ;S ;T ;S ;T ;S ;T ;S ;T ;S ;T ;S ;T ;S ;[CR]
T ;S ;T ;S ;T ;S ;T ;S ;T ;S ;T ;S ;T ;S ;T ;S ;[CR]
COMMENT; RUO message if needed
END_RESULT; checksum value [CR]

7.2.2.3.1 Fields details

Sampling mode for Mythic22AL : PREL ; p

P value	**signification**
"OV"	Prélèvement "Other vial "
"CT"	Prélèvement "Close tube"

Cycle option for Mythic22AL : CYCLE ; o

O value	**signification**
"N"	Used cycle is standard
"A"	Used cycle is optimised for high altitude.

INFO field for Mythic22AL : INFO ; x;y

filed	**value**

x	Nothing or M for order manually matched
y	Nothing or R for rerun.

Alarms and pathological messages :
x : Nothing or alarm.
y : Nothing or interpretive message.

Histograms and thresholds :
S1 S2 S3 : curves thresholds (in [0 et 127] range). S3 is only avalaible for a Mythic18 (LMG). If RUO is not wanted, then PCT and PDW parameters are not transmitted.

m : compressed ASCII field, 1 to 4 characters. Each line size is below 250 characters. Lines number is variable. See below for more details **T** : Matrix Threshold name. See below for more details. **S** : Matrix Threshold value. See below for more details.

7.2.2.4 Haematological parameter format:

WBC; 11.0 ;A ;B ; 2.0 ; 4.0 ; 11.0 ; 15.0 [CR]
Alias ; value ; flag A ; flag B low panic ;low ; high ;high panic [CR]

rank	**Fields**	**Possible Values**
1	Parameter id	WBC
2	Parameter value in the selected unit	5.84 +++++ (5 plus) if over range ..... (5 points) if unvalid result
3	Flag A	Nothing or '*' if result is rejected (counting or measure rejection) or invalidating fault. 's' for dif alarms (see flagging scheme)
4	Flag B	Nothing D : Higher than reportable limits (1) 'L' Result < panic value 'l' Result < normal value 'h' Result > normal value 'H' Result > panic value
5	Low Panic value	2.0
6	Low value	4.0
7	High value	6.0
8	High panic value	8.0

7.2.2.4.1 Alarms list

S-UP NOT DONE
S-UP FAIL
QC NOT DONE
QC FAIL
INS-M
INS-T
INS-P
INS-R
INS-H
L1
L2
L3
L4
L5
R1
R2
P1
P2
P3
NH
NL
RL
N1
N2
IC
HL
W-CL
R-CL
O-CL
O-DF

7.2.2.4.2 WBC interpretive message

LEU>
LEU<
LYM>
LYM<
MON>
NEU>
NEU<
EOS>
BAS>
MYEL
LIC

ALY
ERYB
BLST

NO_INTERPRETATION

7.2.2.4.3 RBC interpretive message

ANE
ERY>
MICRO
MACRO
MICR>
MICR>>
MICR>>>
MACR>
MACR>>
MACR>>>
ANIS>
ANIS>>
ANIS>>>
HYPOCR
COLDAGG
NO_INTERPRETATION

7.2.2.4.4 PLT interpretive report

THR>
THR<
MACROP
PLTAGGR
MICROC
SCHIZ
CELLD
NO_INTERPRETATION

7.2.2.5 Lmne Matrix and threshold details

Lmne matrix is separate into 2 compressed matrixes. This is to provide 2 levels in the third dimension. After uncompressing (using the same owner algorithm), the matrix mean:

- LMNE MATRIX = 2D image for all cellular elements, 1 bit per cell, 128 x 128 bits = 2048 bytes.
- LMNE SHADE MATRIX = only low level cells, 1 bit per cell , 128 x 128 bits = 2048 bytes.
  Low level cells means less than 2 cells.

LMNE MATRIX, bit i	LMNE SHADE MATRIX, bit i	Points ou particules
0	0	absence
0	1	low level
1	0	high level
1	1	Cas impossible

Compressing algorithm overview :

- 00 repeat bytes are coded « Zxxx », xxx = ASCII Hexadecimal counter.
- FF repeat bytes are coded « Uxxx », xxx = ASCII Hexadecimal counter.
- Others bytes are coded in ASCII Hexadecimal: « xx »,.
- Matrix end is coded with « T ».
- Each item is separated by a separator ';'.
- The datas are divided in line lower than 250 characters: a carriage return [CR] is inserted.

Matrix uncompressing function example (C Language):

```
int UnSqueezeMatrix(unsigned char *pucDestMatrix, unsigned char *pucSrcM)  // dest, src
{
int iCnt;
unsigned char *pucDest, *pucSrc, ucVal;
pucSrc  = pucSrcM;     // source
pucDest = pucDestMatrix;   // dest
memset(pucDestMatrix, 0, sizeof(BMP_MATRIX));    // maz dest
while (*pucSrc != 'T')    // end of trame
   {
   ucVal = *pucSrc;
   switch(ucVal)
      {
      case '\x0d':  // ignore CR , LF , separator
      case '\x0a':
      case ';':
         break;
      case 'U':     // value 0xFF
         {
         pucSrc++;
         sscanf((char *)pucSrc,"%x", &iCnt);
         memset(pucDest, 0xFF, iCnt);
         pucDest+= iCnt;
         break;
         }
      case 'Z':     // value 0
         {
         pucSrc++;
         sscanf((char *)pucSrc,"%x", &iCnt);
         memset(pucDest, 0x00, iCnt);
         pucDest+= iCnt;
         break;
         }
      default:
         {
         sscanf((char *)pucSrc,"%x", &iCnt);
```



```
          *pucDest++ = (u_char)iCnt;
          break;
          }
       }
    while (*pucSrc != ';')        // skip ; separator
       pucSrc++;
    pucSrc++;
    }
return((int)pucDest - (int)pucDestMatrix);
}
```

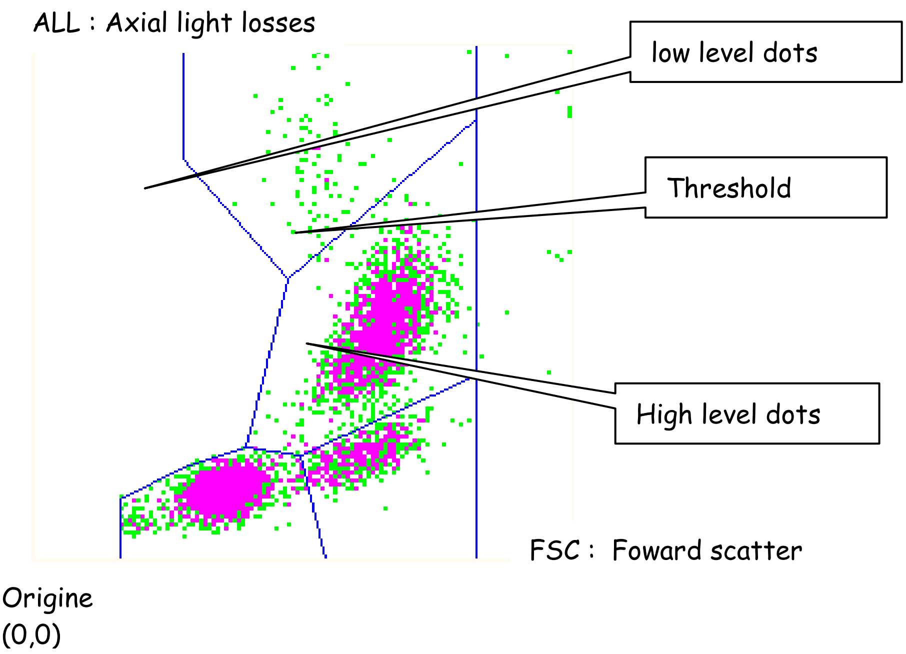
This function needs that « ; » is a separator for scanf() function , usually true for the most systems.

Graphical representation of the matrix :

Coming with the 2-level dot, Mythic22AL applies a color depending of the cell population.

- Low level dot : clear color
- High level dot: darker color

With no population coloring, a simplified representation would give :



Considering a blank bitmap with (0,0) for origin, in the upper left corner, a matrix rendering could be done by the following example (C language)

```
static u_char ucMask[] = {0x80,0x40,0x20,0x10,0x08,0x04,0x02,0x01};
UnSqueezeMatrix( pucMat,  bufferM1);     // dest, src  uncompress
UnSqueezeMatrix( pucMatL, bufferM2);     // dest, src  uncompress
```



```
for (i = 0; i <(16*128); i++)       // 128 x 128 pixels
   {
   ucVal  = *(pucMat+i);        // LMNE MATRIX  byte
   ucValL = *(pucMatL+i);      // LMNE SHADE MATRIX byte
   iX = (i%16)*8;
   iY = i/16;
   for (k=0; k<8; k++)    // MSB to LSB
      {
      if ((ucMask[k] & ucValL) != 0)    // low level
         img->Canvas->Pixels[iX+k]  [iY]   = ColorLow;
      else if ((ucMask[k] & ucVal) != 0)    // high level
         img->Canvas->Pixels[ iX+k]  [iY]    = ColorHigh;
      }
   }
```

The dot (0,0) in the graphical representation is available in [127][0] for uncompressed matrix (128 bytes for ALL, 8 bytes for FSC)

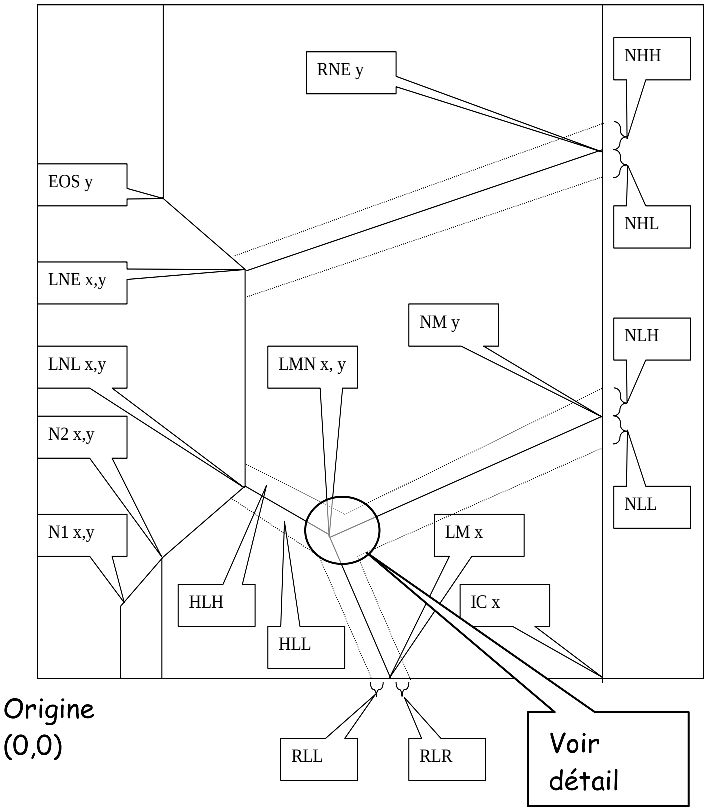
Matrix Thresholds:

The thresholds are on 2 lines after the token «THRES 5D LMNE MATRIX ; «.
They are preceed by their names and provide the X,Y coordinates segment of matrix : the origin is left, bottom, channels 0.

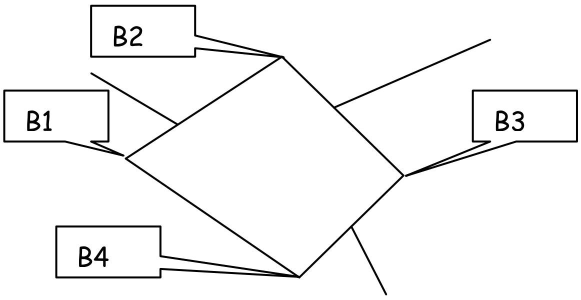
Name	Description	Range mini / Maxi	Default value
N1X	Threshold N1X	0 / 127	20
N1Y	Threshold N1Y	0 / 127	15
N2X	Threshold N2X	0 / 127	35
N2Y	Threshold N2Y	0 / 127	23
LnlX	Threshold LNLX	0 / 127	50
LnlY	Threshold LNLY	0 / 127	28
LmnX	Threshold LMNX	0 / 127	63
LmnY	Threshold LMNY	0 / 127	26
LneX	Threshold LNEX	0 / 127	55
LneY	Threshold LNEY	0 / 127	60
EosY	Threshold EOSY	0 / 127	90
LmX	Threshold LMX	0 / 127	69
ICX	Threshold ICX	0 / 127	105
NmY	Threshold NMY	0 / 127	50
RneY	Threshold RNEY	0 / 127	110
BasoX1	Threshold s B1X	0 / 127	50

BasoY1	Threshold B1Y	0 / 127	25
BasoX2	Threshold B2X	0 / 127	52
BasoY2	Threshold B2Y	0 / 127	33
BasoX3	Threshold B3X	0 / 127	68
BasoY3	Threshold B3Y	0 / 127	29
BasoX4	Threshold B4X	0 / 127	65
BasoY4	Threshold B4Y	0 / 127	22
NHL	Threshold NHH	0 / 127	2
NHH	Threshold NHH	0 / 127	2
NLL	Threshold NLL	0 / 127	2
NLH	Threshold NLH	0 / 127	2
RLL	Threshold RLL	0 / 127	2
RLR	Threshold RLR	0 / 127	3
HLL	Threshold HLL	0 / 127	2
HLH	Threshold HLH	0 / 127	2

Threshold positions (without baso) over the matrix:



Baso detail position :



Matrix and thresholds sample frame:

…………..
LMNE MATRIX;
Z105;10;Z70;4;Zf;8;Z2f;2;Z2f;20;Z2;40;……….;1;0;8;1;80;Zc;8;Ze;2;2;84;Zc;20;0;40;Zf;4;88;Zd;1;20;Zd;
44;Ze;1;18;Z3;20;Z9;1;1;Z2;10;Zc;1;50;Zd;…………………………………………………………………;10;5;Zf;24;20;2;Z3;
Z2;36;70;92;Za;4;Z2;4;a8;6;20;Zc;………………9;40;Zd;a;25;e4;Zb;4;0;25;fe;Zc;4;1;c0;b9;80;Zc;8;99;84;Za;
e;66;38;a0;Zb;1;1a;6d;1c;2;Zb;11;18;…………………;46;36;c4;10;Za;8;9;24;b3;c0;2;Zb;4;a3;e7;5;Z2;2;Z9;
Z3;c;5d;cc;Z8;1;82;40;Z2;84;60;e8;Z8;………………;a4;Z2;10;95;28;4c;Z8;7;8b;33;30;1;0;76;4;Z9;65;0;6c;2;
c0;2b;5a;30;Zc;d1;60;Zd;1;12;8;Ze;92;20;Ze;5b;Z1f;20;Z8c;T;
LMNE SHADE MATRIX;
Z105;10;Z70;4;Z72;40;Zc;10;2;…………………………………………………………………………………;80;Zb;2;Z10;80;
;0;80;Za;20;Z4;80;Zb;80;Z3;80;Z9;……….;10;Z14;20;2;Zb;20;2;62;50;Zd;2;Zc;4;Z2;4;88;4;Zd;8;2e;
40;23;60;Zc;1;44;14;90;Zb;1;8;……….;1;12;44;8;2;Zb;1;18;92;10;2;Za;10;0;42;20;4;10;Zd;b1;Ze;a2;2;4;Z2;2;
92;Zf;19;Z1f;20;Z8c;T;
THRES 5D LMNE MATRIX;
N1X;20;N1Y;15;N2X;35;N2Y;23;EosY;90;LneX;55;LneY;60;LnlX;50;LnlY;28;RneY;110;NmY;46;LmX;69;LmnX;63;LmnY;26;ICX;105;
BasoX1;60;BasoY1;26;BasoX2;64;BasoY2;30;BasoX3;70;BasoY3;29;BasoX4;65;BasoY4;20;NHH;2;NHL;2;RLL;2;RLR;3;NLH;2;NLL;2;HLH;
2;HLL;2;
……………..

7.2.2.6 Result acknowledgement

ACK_RESULT;A;B[CR]
Where A: OK or Error code, B: empty or XB or WG or WG rule numer (QC)

7.2.2.7 Next tube acknowledgement

Mythic22AL acknowledges :

(SEND)
22AL X ;Y ;ACK_NEXT_FOLDER [CR]

7.2.3 CALIBRATION

The calibration frame contains informations related to the calibration :

(SEND)
22AL 1;LOG;**CALIBRATION** en-tête

CALIBRATION	frame tag CALIBRATION
DATE;21/11/2006	date of calibration
TIME;09:25:01	time of calibration
OPERATOR;FDR	calibration operator
ITEM_CALI;6	number of results for the calibration
COEFF	
WBC;1.110	WBC coeff.
RBC;1.220	
HGB;1.440	
HCT;1.550	
PLT;1.330	
TARGET	frame tag TARGET
UNIT;1	
LOT;TI855X	lot (max 8 chars)
LOT DATE;20/11/2006	creation date of the target
LOT TIME;14:32:20	creation time of the target
EXPIRY DATE;03/10/2016	expiry date of the target
OPERATOR;BH2	target operator
WBC;6.3 ;1.3	target value; tolerance in the specified unit
RBC;4.33 ;1.83	
HGB;13.3;2.1	
HCT;43.3;5.3	
PLT;253 ;33	
END_CALI; 55360	end of frame; CRC control

This frame is composed with : header + CALIBRATION frame + TARGET frame + end of frame
In the case of a direct coeff. input, lot informations are blank and the number of results is zero.
The acknowledgment is done with the following frame :

ACK_CALI ;Lot;A [CR]
Où A : OK ou Code d'erreur.
Where A : OK or Error code.

7.2.3.1 Target calibration loading

In order to modify calibration target of Mythic22AL:
(RECEIVE)
SET_CALIBRATION_LOT [CR]
Mythic22AL acknowledges the request :
(SEND)

22AL X ;Y ;ACK_ SET_CALIBRATION_LOT ;A [CR]
Où A : OK ou Code d'erreur BUSY.
Where A :OK
Trame de description de la cible :
(RECEIVE)

TARGET	frame tag
UNIT;1	
LOT;TI855X	lot
LOT DATE;20/11/2006	(OPT)creation date of the target
LOT TIME;14:32:20	(OPT)time creation of the target
EXPIRY DATE;03/10/2007	expiry date of the target
OPERATOR;BH2	(OPT)target operator
WBC;6.3 ;1.3	target value; tolerance in the specified unit
RBC;4.33 ;1.83	
HGB;13.3;2.1	
HCT;43.3;5.3	
PLT;253 ;33	
END_CALIBRATION_LOT; 5577	frame tag; CRC control
[**ETX**]	caractère fin de bloc (0x03)

Except line **END_CALIBRATION_LOT** and **ETX,** this frame **TARGET** is identical to the one included in the CALIBRATION frame above.

Lot reception acknowledgment
(SEND)
22AL X ;Y ;ACK_CALIBRATION_LOT; A [CR]
Où A : OK ou Code d'erreur BUSY.

7.2.3.2 Calibration results

Results calibration are sent in block :
(SEND)

22AL X ;Y ;**RESULT**	en-tête
DATE;30/10/2003	
TIME;15:36:38	
MODE;**CALIBRATION**	
UNIT ; Unité	
SEQ;1; 0	Order number 1 to n ; 0 (not used)
TEST;LMG	
OPERATOR ; JDF	operator who did the analysis
PREL;CT [CR]	
CYCLE;A [CR]	
WBC; 11.0; A;B	parameter value and its flags A and B, idem MODE NORMALresult
RBC; 6.00; A;B	
HGB;15.0; A;B	
HCT;49.0; A;B	
PLT; 320; A;B	
END_RESULT; 2274	end of frame; CRC control

… followed by a similar block for the next result.

To sum up, at the end of a calibration, Mythic22AL send a CALIBRATION frame (including a TARGET frame) then RESULT frame in CALIBRATION mode.
This is the line CALI_ITEM of the CALIBRATION frame which gives the number of results.

7.2.4 QC

7.2.4.1 Target QC loading

In order to modify a QC target :
(RECEIVE)
SET_QC_LOT; A[CR]
Where A is the target rank on the instrument (1 to 12)

Mythic22AL acknowledges the request :
(SEND)
22AL X ;Y ;ACK_ SET_QC_LOT ;A [CR]
Where A : OK or Error code BUSY :

The QC target frame looks like the one of a CALIBRATION target. It contains the Level field and some extra hematologic parameters.
(RECEIVE)

TARGET	frame tag
UNIT;1	
LOT;KDF9547	lot
LOT DATE;20/11/2006	(OPT)creation date of the target
LOT TIME;14:32:20	(OPT)creation time of the target
LEVEL;H	target level high, medium, low ('H', 'N' or 'L')
TEST;DIF	CBC or DIFF
EXPIRY DATE;03/10/2007	expiry date of the target
OPERATOR;BH2	(OPT)target operator
WBC;5.1 ;1.1	target value; tolerance in specified unit
RBC;4.61 ;0.81	
HGB;12.1;1.1	
HCT;43.1;1.3	
PLT;221 ;11	
LYM;2.1 ;0.8	
MON;0.8 ;0.4	
LYM%;30.5;1.1	
MON%;13.0;1.5	
MCV; 81.7;1.1	
MCH; 25.0;1.2	
MCHC; 30.6;1.2	
RDW; 8.1;1.5	
MPV; 7.5;0.7	
PCT; 0.401;0.055	

PDW; 9.9;1.3
EOS;1.4;0.4
BAS;0.6;0.3
EOS%;2.4;0.4
BAS%;1.6;0.3

END_QC_LOT; 2547	frame tag; CRC control
[**ETX**]	end of block char (0x03)

Aknowledgment of the lot
22AL X ;Y ;ACK_QC_LOT ; A [CR]
Where A : OK or Error code

7.2.4.2 QC Result

A result may be sent at the end of an analysis, it also contains informations about QC target.

(SEND)

22AL X ;Y ;**RESULT**	header
DATE;30/10/2003	analysis date
TIME;15:36:38	time analysis
MODE;QC	
UNIT ; 1	
SEQ;352	Sequence number of the analysis (*)
LOT ; KDF875	lot number
LOT DATE;20/11/2006	creation date of the target
LOT TIME;14:32:20	creation time of the target
LEVEL ;N	target level high, medium, low ('H', 'N' or 'L')
EXPIRY DATE; 03/10/2007	expiry date of the target
USER;BH2	logged operator
TEST;DIF	CBC or DIFF
OPERATOR ; BYLON	target operator
PREL;CT [CR]	
CYCLE;A [CR]	
WBC;11.0 ;;H;4.0 ;6.2	(parameter; flag A; flag B ; low target value ; high target value)

RBC;4.10 ;;;3.80 ;5.42
HGB;10.8;;L;11.0;13.2
HCT;34.8;;L;41.8;44.4
PLT;120 ;;L;210 ;232
LYM;2.7 ;;;1.3 ;2.9
MON;2.8 ;;H;0.4 ;1.2
LYM%;22.0.8 ;;L;30.5;31.1
MON%;12.8 ;;L;13.0;13.5
MCV; 82.8 ;;;81.7;91.1
MCH; 32.4 ;;H;25.0;27.2
MCHC; 30.8 ;;;30.6;31.2
RDW; 8.8 ;;;8.1;10.5
MPV; 7.9 ;;;7.5;8.7

PCT; 0.452 ;;;0.401;0.555
PDW; 10.1 ;;;9.9;10.3
EOS; 2.8 ;;H;1.4;2.4
BAS; 2.7 ;;H;0.6;1.3
EOS%;2.8 ;;;2.4;3.4
BAS%;3.2 ;;H;1.6;2.3
END_RESULT;959 end of frame; CRC control

(*) right after the analysis, transmitted sequence number is the one applied by the Mythic22AL. For a post-send, the number is between 1 to n.

7.2.5 REPEATABILITY

7.2.5.1 Result

A result may be sent at the end of an analysis :

(SEND)	
22AL X ;Y ;**RESULT**	Header
DATE;30/10/2003	date of the analysis
TIME;15:36:38	time of the analysis
MODE; REPEATABILITY	
UNIT ; 1	
SEQ;352	Sequence number of the analysis (*)
TEST ; DIF	
OPERATOR ; BAB	logged operator.
PREL;CT [CR]	
CYCLE;A [CR]	
WBC;11.0 ;;H	(parameter value ;flag A ;flag B)
RBC;4.10 ;;	
HGB;10.8;;L	
HCT;34.8;;L	
PLT;120 ;;L	
LYM;2.7 ;;	
MON;2.8 ;;H	
LYM%;22.0.8 ;;L	
MON%;12.8 ;;L	
MCV; 82.8 ;;	
MCH; 32.4 ;;H	
MCHC; 30.8 ;;	
RDW; 8.8 ;;	
MPV; 7.9 ;;	
PCT; 0.452 ;;	
PDW; 10.1 ;;	
EOS; 2.8 ;;H	

BAS; 2.7 ;;H
EOS%;2.8 ;;
BAS%;3.2 ;;H
END_RESULT;3259 end of frame; CRC control

(*) right after the analysis, transmitted sequence number is the one applied by the Mythic22AL. For a post-send, the number is between 1 to n.

7.2.6 STARTUP frame

This frame is sent after a STARTUP cycle completion :
(SEND)
22AL X; Y; **STARTUP**; Date; Time; Status; channel value 1; channel value 2; channel value 3; channel value 4 [CR]

Date : jj/mm/aaaa
Time : hh:mn:ss
Status : PASS, FAIL

7.2.7 REAGENT frame

When a reagent is being changed, the following frame is sent :
(SEND)
22AL X; Y; **REAGENT**; Date; Time; Type; N° Lot; Expiry Date; Volume[CR]

Date : jj/mm/aaaa
Time : hh:mn:ss
Type : texte
N° Lot : texte
Expiry Date: jj/mm/aaaa
Volume : full

Type : DILUENT, CLEANER ou LYSIS.

7.3 CRC checksum

The sum of control is a cyclic redundancy code (CRC-16)

7.3.1 Algorithmic

The code used is supplied below (C language), it allows computing of a CRC-16 value
This algorithm is mainly used on embedded systems. For better performance, compute is based upon an array containing 16 values, which gives the following CRC:

```
CRC = 0xFFFF
For each nibble :
```



```
Work on high weight nibble :
Index = byte EXCLUSIVE OR CRC
Index = Index AND 000F
CRC = Table(Index) EXCLUSIVE OR (CRC divided by 16)
Work on low weight nibble :
Index = byte divided by 16
Index = Index EXCLUSIVE OR CRC
Index = Index AND 000F
CRC = Table(Index) EXCLUSIVE OR (CRC divided by 16)
```

<u>C language source example implementation :</u>

Seek table declaration :

```
static const unsigned short ausCrcTab1[] =
 {
 0x0000, 0xCC01,0xD801,0x1400,0xF001,0x3C00,0x2800,0xE401,
 0xA001,0x6C00,0x7800,0xB401,0x5000,0x9C01,0x8801,0x4400,
};
```

CRC computation :

```
unsigned short calc_crc(unsigned char *pucData, long lSize)
{
unsigned short usAcc1 = 0xFFFF;

 while ( lSize > 0 )
      {
      /* gestion par quartet du calcul  */
      usAcc1 = ausCrcTab1[(*pucData ^ usAcc1) & 15] ^ (usAcc1 >> 4);
      usAcc1 = ausCrcTab1[((*pucData >> 4) ^ usAcc1) & 15] ^ (usAcc1 >> 4);

      pucData++;
      lSize--;
      }

return(usAcc1);
}
```

^ : ou exclusif.

>> : décalage logique à droite.

& : et logique.

CRC value is computed from the beginning of the emission to the end of line [CR] just before the checksum line (title + value)

8 RECEIVING A WORKLIST FROM A LIS

8.1 Principles

A LIS may connect in HOST (either NET or RS), in order to send patient files to an AL22. Each send is checked by the AL22 (CRC based) and acknowledged by such. If errors occur, they are sent back to the LIS for processing.

A LIS may transmit patient files using the "ADD_NEW_ORDER" command then processed by the AL22, as defined below:

ADD_NEW_ORDER,KEY,RTYPE,RACK,POS,Stat,SID,PID,ID,BIRTH,SEX,TYPE,TEST,PRESC,LOCAT,PRELD, PRELTIME,RUN,MATCH,Pat.COMMENT,CRC[CR]

The ADD_NEW_ORDER command is ascii-based and must be *carriage-return* terminated.
Below the parameters to be supplied:

KEY:	File number. AL22 will not take this value into account, since it assigns a new file number for the received order. Remains at 0.
RTYPE:	Rack Type number, must be : {undef, 1, .., 10 }
RACK:	Rack number, must be : {undef, 1, .., 10 }
POS:	Position number for the sample inside a rack, must be: {undef, 1, .., 5 }
Stat:	Order Status. AL22 will not take this value into account since a received order is considered to be to do. Remains at 'T' for TODO.
SID:	Order SID, 16 chars max, **mandatory**
PID:	Order PID, 16 chars max.
ID:	Order ID, 20 chars max.
BIRTH:	Patient birthday with the following format **DD/MM/YYYY**
SEX:	Patient Sex, must be: {0, 1, 2} respectively : '**undef**', '**Masculine**', '**Féminine**'
TYPE:	Blood Type, 10 chars max. **STANDARD** for instance.
TEST:	Test Type, must be: {0, 1} respectively, '**CBC**' , '**DIF**',
PRESC:	Physician Name, 20 chars max.
LOCAT:	Examination location, 20 chars max.
PRELD:	Sampling Day, must be {0,1,2} respectively '**UNKNOWN**' , '**TODAY**', '**YESTERDAY**'
PRELTIME:	Sampling Hour with the following format **HH:MM:SS**
RUN:	Remains blank.
MATCH:	Remains blank.
Pat.COMMENT:	Optional Comment, 30 chars max.
CRC:	unsigned short value containing the CRC

Note : Only SID and CRC fieds have to be provided for the order to apply, as long as commas are provided even in case of blank fields.

Below, an example of an order supplied to an AL22. This order concerns a sample laid inside the 2nd slot of the first rack of type 3. The patient was born in 1990 and the sample was achieved by doc. House in the state of Oregon.

```
ADD_NEW_ORDER,0,3,1,2,T,TEST SID 1,TEST PID 1,TEST
ID,01/01/1990,1,STANDARD,1,HOUSE,OREGON,2,00:00:00,,,comment,6410\r
```

8.2 CRC value

CRC computing (6410 in the previous example) is done from the first comma (excluded) to the last comma (excluded). In other words, CRC is computed using the string in yellow background.

The following CRC computing is applied :

Search array for performance efficiency

```
static const unsigned short ausCrcTab1[] =
 {
 0x0000, 0xCC01,0xD801,0x1400,0xF001,0x3C00,0x2800,0xE401,
 0xA001,0x6C00,0x7800,0xB401,0x5000,0x9C01,0x8801,0x4400,
};
```

CRC computing :

```
unsigned short calc_crc(unsigned char *pucData, long lSize)
{
unsigned short usAcc1 = 0xFFFF;

 while ( lSize > 0 )
      {
      usAcc1 = ausCrcTab1[(*pucData ^ usAcc1) & 15] ^ (usAcc1 >> 4);
      usAcc1 = ausCrcTab1[((*pucData >> 4) ^ usAcc1) & 15] ^ (usAcc1 >> 4);

      pucData++;
      lSize--;
      }

return(usAcc1);
}
```

8.3 FILE receipt ACKNOWLEDGEMENT

Receipt acknowledgement is done after CRC check. After a proper transmission and order processing, AL22 sends back the following line to tell the LIS that no error has occurred.

ADD_NEW_ORDER: 0, OK[CR]

In case of errors, AL22 may send back the previous command to LIS, with the error number followed by one of the following string:

ERR_WL_IS_FULL,	Worklist is full
ERR_WL_ORDER_IS_USED,	Can't modify file because the status is not TODO.

ERR_WL_ORDER_SID_IS_BLANK,	Empty SID is not allowed
ERR_WL_ORDER_RACKPOS_USED,	Rack/Position already used within the worklist.
ERR_WL_ORDER_PID_IS_BLANK,	Depending of the SETUP, file must contain a PID
ERR_WL_ORDER_ID_IS_BLANK	Depending of the SETUP, file must contain a ID
ERR_WL_ORDER_BAD_CRC	CRC error.

Those strings may be parsed to detect errors and decide whether resume order sending or not. An order successfully sent to the AL22 will not be compromised by another order which leads to an error.